Validation of the Proposed Round 10 Young-Tier Answer

Independent shape check and exact source-target product enumeration. This report validates the uploaded output; it does not introduce a new approximation.

Verdict

The uploaded answer is not valid as written. Its final 49-55 u claim follows from dimension/FLOP errors and from an unsupported shared-suffix saving. The corrected exact count for the young-tier square-product core is 193 n x n products. Under the prompt's L5 Strassen billing (0.51 u per eligible square product), that core is 98.43 u, before small elementwise/packing work. This is consistent with the measured approximately 104 u young-tier ledger and is nowhere near 55 u.

1. Exact count I ran

I enumerated target layers 2..16 with young ages 1..4. Full D21 is needed through layer 15; the final layer needs only d3, so its dense D21 products are omitted. Age-1 P is the current map M and does not require a separate P product. Every A use still requires one dense application/transport.

Quantity	Exact count	Reason
Young source-target uses through L16	54	sum_l min(4,l-1), l=2..16
P map/update products	39	only ages 2-4 need an explicit P product
A application/transport products	54	one for every young source-target use
D21 dense contractions	100	2 per pair, 50 pairs through L15
TOTAL square products	193	39 + 54 + 100
L5-Strassen core cost	98.43 u	193 x 0.51 u

Independent enumeration output: P = 39, A = 54, D21 = 100, total = 193. This was computed in this session from the layer/age schedule, not inferred from the approximately 104 u measurement.

2. Specific errors in the uploaded answer

Row-stacked transport is mispriced.
For Z=[P;A] in R^(2n x n), the written block operator (I2 kron M) is 2n x 2n. A generic multiply costs 2*(2n)*(2n)*n = 8n^3 FLOPs = 4 u. Exploiting the block-diagonal structure reduces it to two independent n x n products = 2 u, not 1 u. Therefore stacking P and A does not halve transport.

Four-age batching has the wrong real shape.
For one D21 term, four age blocks concatenate as n x 4n multiplied by 4n x n. That costs 2*n*(4n)*n = 8n^3 FLOPs = 4 u, exactly the cost of four separate square products. There are two D21 terms, so batching is FLOP-neutral.

The 6.5 effective layers normalization is circular.
It is not a structural count. The exact combinatorics are 50 D21 source-target pairs through L15 and 54 young source-target state uses through L16. Using 104/16=6.5 and then multiplying per-layer costs by 6.5 creates the contradictory 52 u / 104 u passages in the uploaded answer.

The shared-suffix saving is not established.
Precomputing a consecutive-map suffix requires a dense map product; applying that suffix to a source-specific birth matrix/leg requires another dense product. The exact hierarchical schedule therefore pays 39 interval-map products + 54 birth applications + 100 contractions = 193 products - exactly the forward count.

The final 5.55 u/layer and 49-55 u claims are unsupported.
They depend on subtracting an unproven suffix reuse and on the incorrect stacked-transport arithmetic. They also conflict with the answer's own final paragraph, which says the young tier remains about 86 u.

3. Current order versus the best exact hierarchical order

The most favorable exact alternative is to store consecutive interval maps and reconstruct A only when a birth-target pair is queried. This is the natural mixed/hierarchical competitor because it maximizes reuse of the M chain without approximating any source.

Exact order	P / interval	A / birth	D21	Total
Optimized forward	39	54	100	193
Hierarchical interval-map	39	54	100	193
Four-age wide batching	same FLOPs	same FLOPs	same FLOPs	193 equivalent

Why the tie is structural: the Hadamard nodes P o P and P o A bind two entries carrying the same row index before the trailing source-specific leg can be contracted. Summing source identities before this nonlinear node would create cross-source terms that are not present in D21. Keeping source identities separate until after the Hadamard node leaves one dense trailing contraction per term and per source-use. Reassociating the linear M chain can move where the products are paid, but not remove them.

4. What is and is not validated

Validated: batching is FLOP-neutral; row-stacking P/A is FLOP-neutral; exact suffix/interval reuse ties the forward count; the square-product core is 193 products under the stated layer schedule.

Not validated from the uploaded HTML alone: the claim that P^-1 A drifts by 100-970 percent. The attachment cites that as a measurement but provides no underlying capture output in the document. It should not be used as a premise unless separately verified from the stored arrays.

Also not validated: any claim that applying Strassen-5 only to transports newly saves a specific number of u. Whether that is new depends on what the approximately 54 u measured transport figure already includes. The safe statement is only the exact algebraic core count above.

5. Corrected bottom line

Do not implement the uploaded shared-suffix / stacked-pair method. Its claimed 55 u result is an arithmetic artifact. Under exact dense young-source algebra and the clarified final-layer trim, the best contraction orders checked have the same 193 square-product core. At 0.51 u per square product this is 98.43 u, before small overhead. Therefore a <=55 u exact young tier requires a genuine change of representation or a new algebraic identity that removes source-specific dense products - not a different contraction order.

Recommended falsifier for any future exact-reuse claim: list every GEMM with dimensions (m,k)x(k,p), compute 2mkp FLOPs, and compare the complete schedule against the 193-product reference. Any claimed saving must reduce that exact sum while reproducing D21 to <=1e-6 on stored dense legs.